

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456337.6788 +/- 0.00047 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1056 +/- 0.0093
Transit depth (Rp/Rs)^2	1.11 +/- 0.2 [%]
Orbital Inclination (inc)	86.98 +/- 1.1
Airmass coefficient 1 (a1)	1.2674 +/- 0.0079
Airmass coefficient 2 (a2)	-0.1169 +/- 0.0025
Scatter in the residuals of the lightcurve fit is	0.61 %
Best Comparison Star	None
Optimal Aperture	4.46
Optimal Annulus	11.15
Transit Duration (day)	0.121 +/- 0.0091

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1056 ± 0.0093 vs 0.1241 (deviation 1.2σ)
Duration fit vs published	0.121 d vs 0.1317 d (deviation 0.71σ)
Mid-time O–C	-1.3 min
Depth SNR	6.9
Residual scatter	0.61 %

Light curve

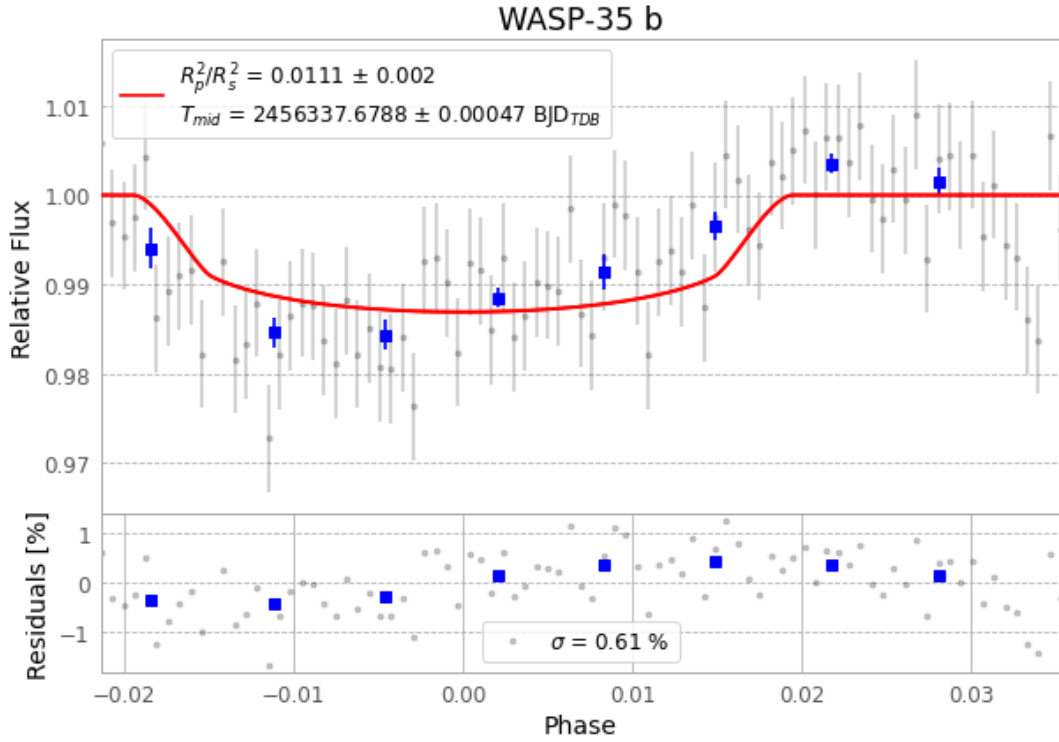


Figure 1 — FinalLightCurve_WASP-35 b_2013-02-13.png (SHA-256 a48e1cef150ea6a4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [318, 322], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 47893afda94c2371...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

87 FITS frames (57 MB), WASP-35130214023612.FITS → WASP-35130214065412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-130214.log (4e11cc2afa87...), FinalLightCurve_WASP-35 b_2013-02-13.png (a48e1cef150e...), FinalLightCurve_WASP-35 b_2013-02-13.csv (aa464bda6ebc...)

Compute resources

Wall time 175.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 02:45Z.